The World Economy over the Very Long Run

A History in Data and Debates

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Preface

This is a history of the world economy over roughly five thousand years — from the first riverine cities to the container ship — written from an unusual vantage point. Most accounts of the long-run world economy are told, implicitly, from the North Atlantic looking outward: Europe rises, the world responds. This book stands instead on the shore of the **Indian Ocean** and looks the other way. From there a different shape appears. For most of recorded history the centre of gravity of the world economy sat in **Asia** — visible in the direction precious metal flowed and in where the world's goods were actually made — and Europe's two centuries of dominance look less like the natural order of things than like a real, transformative, but **bounded** interruption.

That is a claim, not a creed. The book's method is to **test it**, period by period, against the evidence — and to be honest about where the evidence runs thin or the scholars disagree. Hence the subtitle: *a history in **data and debates***.

How the book works

Every chapter asks the same **four questions** of its period — about the *direction* of integration, the *channels* that carried it, the *modes* (goods, technology, people, capital) that did the integrating, and the changing *role of Europe*. Those four questions are the reader's toolkit; they are introduced in full in the Introduction.

Every chapter also has the same recurring furniture, designed so the book can be read as a narrative *or* mined as a course text:

- a **Data exhibit** — one empirical centrepiece, with its provenance and its caveats;

- a **Debate** — the live scholarly controversy, presented as genuinely open;
- running **callouts** that *follow the money* and ask *how we know*;
- a **source endnote after every paragraph** — pointing you to the evidence behind the claim and to where to *read more*, so the prose stays clean but nothing is unsourced;
- a short **"At the research frontier"** review of recent quantitative (cliometric) work on the period — the live edge of the scholarship, not just the classics;
- and end-of-chapter **questions for consideration** (3–5, essay/exam style), a reproducible **data exercise**, and **further reading** — with a bank of **cross-cutting questions** at the end of the book.

The narrative between that furniture is free to breathe — some periods leave abundant data, others almost none — but the scaffolding is constant, so you always know where you are.

Open access

This book is published open-access online and as a downloadable PDF. Its data exercises are **reproducible**: the figures and tables are generated from datasets you can download and re-run yourself. Where a number is disputed, the book says so and shows you the dispute, rather than quietly picking a side.

> **i** Draft status
>
> This is an in-progress manuscript. The Introduction and Chapter 2 are first-draft pilots that set the template; the remaining chapters are being drafted from the module's design corpus. See `world_economy_books/textbook_plan.md` for the build plan.

Introduction: The vantage, the questions, the method

A claim to test, not a creed to defend.

The view from the other shore

Imagine the world economy not from London or New York but from a beach on the Malabar coast, or the deck of a dhow running before the monsoon from Aden to Gujarat. It is a deliberately awkward place to stand, because almost every long-run history of the world economy is written from somewhere else — from the North Atlantic, looking outward, with Europe as the active party and the rest of the world as the stage on which European energy plays out. Tell the story that way and its shape is settled in advance: the world is poor and static until, around 1500 or 1750, Europe stirs, and modern prosperity radiates from a single source.

Stand on the Indian Ocean shore instead and a different shape appears. You notice that for most of recorded history the largest and richest economies, and the most commercially developed, were in Asia — Han and Song China, the Abbasid caliphate, Mughal India — and that the world's most coveted store of value, silver and gold, flowed toward them, not away. Europe, by contrast, spent long centuries as a periphery: an exporter of slaves, furs and raw metal, an importer of manufactures and ideas, paying for what it wanted in bullion it could barely produce itself. And the period in which Europe genuinely ran the world economy is short — concentrated in the century or so after 1750 — and even at its height it rested on the Asian periphery it governed.

This book takes that view seriously enough to build a whole history around it. But the vantage point is a tool, not a slogan.

The argument it generates is a claim, and the book's job is to test that claim, period by period, against what the evidence will bear, and to be candid about where the evidence is thin and where serious scholars disagree.

The spine: the claim we are testing

Here is the argument in one paragraph; the rest of the book interrogates it.

> *Viewed from the Indian Ocean, the world economy spent most of its history centred on Asia — visible in the direction bullion flowed and in where the world's goods were made. Integration advanced and retreated in waves shaped by climate, disease, technology and money, along a durable maritime channel and a periodically resurgent overland one. Europe's dominance was real and transformative but bounded — concentrated in the century after 1750, built partly on the Asian centrality it overturned, and resting at its height on the Indian Ocean periphery it governed. Its timing and depth remain debated, and the East's aggregate return — measurable but partial, achieved within a Western-built order — is the long arc closing, not simply repeating.*

Call this the spine. It treats European dominance as an interruption rather than a destination, but a bounded one. Four qualifications bound the claim, and they recur throughout the book:

1. the interruption was consequential — roughly two transformative centuries, not a blip;
2. its timing and depth are genuinely open, debated in Chapter over the Great Divergence;
3. the eventual return east is partial — aggregate weight, not a restoration of per-capita parity; and
4. it is system-dependent — the Asian re-rise happened within a US-anchored order.

We do not settle the spine here. We assemble the tools to test it, and return to weigh it in the Chapter .

The four questions

The book asks the same four questions of every period. They organise every chapter and are the method the book hands you. Together they answer the spine — *where is the centre of gravity, and is it moving?* — without pre-judging it.

! The four motivating questions

- **Q1 — Direction.** Is the system integrating or disintegrating? (Two refinements the evidence forces on the simple cycle: integration breeds fragility, since the links that carry goods carry shocks; and disintegration has winners, since a fracturing core can loosen its grip on a periphery.)
- **Q2 — Channels.** Which routes carry the integrating weight, maritime or overland, and are they complements or substitutes? (The maritime Indian Ocean is the durable artery; the overland corridor had one true climax, the Mongol Pax of Chapter .)
- **Q3 — Modes.** Which of the four flows — goods, technology, people, capital — is doing the integrating? Here the book makes one move: treat bullion as a *good* until the nineteenth century, then as *money and capital* after, and read its direction as the running tracer of the centre of gravity.
- **Q4 — Europe.** What is Europe's role — peripheral, entrant, dominant, one of several cores — and how does it change? Asked as an open question, not assumed.

The spine is answered jointly by Q3, which follows the bullion and then the capital, and Q4, Europe's changing role; Q1 and Q2 describe how the system that holds the centre is wired at each moment. The arc the four questions trace, chapter by chapter, is this:

> no single centre (Ch 1) → Asia the production-centre of a multipolar relay (Ch 2) → Asia central, Europe peripheral (Ch 3) → Asia central, a first north-west-European seed (Ch 4) → Asia the sil-

ver sink (Ch 5) → Asia the workshop, Europe armed (Ch 6) → the hinge, the centre moves west (Ch 7) → peak Western dominance on an Indian-Ocean base (Ch 8) → the order cracks within the West (Ch 9) → the return toward Asia (Ch 10).

The bullion tracer, and why it flips

Before modern capital markets, metal moved toward the workshop. People who made things the rest of the world wanted — Indian cotton, Chinese silk and porcelain — were paid in silver and gold, which pooled where production was most valued. So in antiquity Roman silver drained *east* to India (Chapter); in the early-modern centuries New World silver flowed *east* to China (Chapter). The tracer even works in reverse where it should: the Abbasid core paid silver *north* to its slave-and-fur periphery (Chapter), mirroring Rome.

After the nineteenth century the logic flips, because bullion's role changes. Under the classical gold standard and integrated capital markets, gold becomes money and capital, and following it leads not to the workshop but to the financial core — London, then New York. But note who underwrote that core: India's colonial "drain" helped finance Britain's payments deficits (Chapter), sourced in the periphery and intermediated by the core.

Six threads that run the length of the book

Beyond the four questions, six threads run through the book; this section sets them out.

1. **Centre of gravity —** the spine itself, tracked with an explicit evidence-ledger entry and a stated confidence level in every chapter's verdict.
2. **Measurement of success: quantity to quality.** This thread corrects a recurring error. Before about 1800, like every other species, humans turned prosperity into numbers — more people, more output, bigger empires (aggregate, *extensive* growth). Only after about 1800 did any society turn it into sustained

income per head (per-capita, *intensive* growth, the escape from the Malthusian ceiling, and a uniquely modern one). The thread is set up in Chapter , runs through the silver story in Chapter and Chapter , and resolves at Chapter , where the Great Divergence *is* the switch. It is what stops us reading every "thriving" empire — Han, Song, Mughal, Qing — as growth in the modern sense.

3. **Follow the silver** — the early-modern form of the bullion tracer (Chapter to Chapter). The same American silver met different monetary conditions and so had different fates: Asia absorbed and specialised, the Ottomans inflated, West Africa debased and depleted, and Europe converted its monetary loss into armed-trade and financial power. That last fate is the bridge to the divergence.
4. **Measurement of integration: how we know.** The evidence for measuring integration improves over time, from the diffusion of goods and ideas, to bullion flows, to wages and prices, to GDP, to a computed centre of gravity. Before prices, only the diffusion prong exists, which limits what Chapter and Chapter can claim.
5. **Forces.** Four drivers sit behind Q1's direction — climate, disease, technology, policy — two of them natural, a reminder that the world economy has always been embedded in a natural system.
6. **Data survival and bias.** What survives reflects record-keeping conventions and the accidents of climate and preservation. Absence of data is not absence of activity, and the surviving record under-documents the very Indian-Ocean world this book centres on. Whose eyes we see through is itself a running question.

A note on method

Two commitments follow, and they are why the subtitle is *data and debates*.

- **Data.** The book leans on numbers — trade volumes, bullion flows, real wages, output shares, the computed centre of gravity — because numbers are how you adjudicate a claim like the spine rather than just asserting it. The figures and tables are reproducible: generated from datasets you can download and re-run. But the data thin out sharply as you go back, and they are biased toward the literate, fiscal, European end of the record. A chapter on the Bronze Age and a chapter on the twentieth century are not equally evidenced, and the book says so.

- **Debates.** Where the scholarship is genuinely contested — when the Great Divergence began and why; whether interwar disintegration was man-made or inevitable; when "globalisation" really started — the book presents the disagreement rather than resolving it quietly. Disputed figures are flagged and taught, not silently resolved. The aim is to hand you the questions, the evidence and the open arguments, so you can test the spine yourself.

i How to read the chapters

Each chapter opens by following one object that carries its period's story, locates the period on the arc, sets out the cast and how we know, narrates the period on its own terms, then reads it through the four questions, delivers a centre-of-gravity verdict with a confidence level, and hands off to the next. Look out for the recurring Data exhibit and Debate, and the Follow the money and How we know callouts.

I — A world without a centre

Deep structures: a world without a centre

Before prices, the only measure of integration is the diffusion of goods and ideas.

Draft pending

Period: c.3500–500 BCE. This chapter is a stub. Draft it from the standardized template (`_chapter-template.qmd`) using its story spine (`teaching/ecu4413_world_econ/module_design/topic_01_story_spine_2026-06-19.md`), the matching topic brief, the centre-of-gravity ledger, the data compendium, and `datasets_by_topic.md`.

The first world system: Rome was the customer, Asia the workshop

For the first time we can ask where the economy's centre lay — and the answer is not Rome.

Follow one thing: a stream of Roman silver running east

Around the middle of the first century CE, the elder Pliny grumbled that the Roman empire was haemorrhaging precious metal to pay for the luxuries of the East — by one of his figures, 100 million sesterces a year drained away to India, China and Arabia. The number is rhetorical and much argued over, but the *direction* is not. Across the early centuries CE, Roman silver and gold flowed steadily eastward across the Indian Ocean and, having arrived, largely stayed: it was melted, re-struck by the Kushan and Saka rulers of north-west India, and absorbed into Asian economies that valued it as metal. Follow that one stream of metal and you have the argument of this chapter. The richest, hungriest consumer of the ancient world paid, in hard coin, an economy whose workshops sat not on the Tiber but on the Tamil coast and beyond. Rome was the customer. Asia was the workshop.[1]

[1] **Sources:** Pliny, *Natural History* 6.101, 12.84, via Dalrymple, *The Golden Road* (2024); McLaughlin, *The Roman Empire and the Indian Ocean* (2014). **Read more:** De Romanis, *The Indo-Roman Pepper Trade and the Muziris Papyrus* (2020) on what the figures can and cannot bear.

Where we are on the arc

In Chapter there was no centre to find — only loosely linked riverine cores and the slow diffusion of metals and crops along three nascent channels. This chapter is where, for the first time, the question *where is the centre of gravity?* can actually be asked, because for the first time the channels fuse into a single, measurable trans-Eurasian relay running from Roman Britain to Han China. The conventional answer is "Rome." This chapter tests that answer by following the goods and the money — and finds that the connected economy's productive core lay in Asia, with Rome as its structurally dependent western terminus.[2]

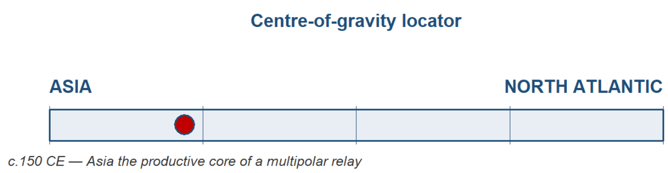

Figure 1: Centre-of-gravity locator (c.150 CE): Asia the production core of a multipolar relay; Rome the western consumer-terminus.

The stage and the cast

The right mental picture is **not** a wheel with Rome at the hub and spokes running out to the provinces and beyond. It is a **triangle** — Rome, India and China — connected by a relay of intermediary empires, each of which took its cut. The main chain ran Rome to Parthia and the Kushans to Han China; Aksum (in the Red Sea), the Arabian incense kingdoms, and the Indian states rounded out the cast. No single merchant or polity ran the system end to end. It was a **relay of empires,**

[2]**Sources:** Findlay & O'Rourke, *Power and Plenty* (2007), ch. 1; the centre-of-gravity ledger, `module_design/centre_of_gravity_ledger.md`. **Read more:** Beaujard, *The Worlds of the Indian Ocean*, vol. 1 (2019).

not one market — a distinction we will need when we ask, later, whether this was really "one" world economy.[3]

The cast of this chapter — Rome, Parthia and the Kushans, Han China, India, and the smaller players around the rim — is profiled below. Click any actor to expand how its economy worked, at home and in the relay. India, present in every chapter of this book, is given the fullest treatment.

Rome — the wealthy consumer

Rich, hungry, and paying in coin. An unusual state for its time: a commercial-fiscal empire whose treasury leaned heavily on the customs of trade, and whose appetite for eastern goods it could satisfy only by exporting bullion it could not replace at home.[4]

Parthia and the Kushans — the middlemen

The intermediary empires that held the overland middle: Parthia astride the routes west of the Euphrates, the Kushans astride the passes of the Hindu Kush and the gateway to Central Asia. They taxed the silk that passed through and re-minted the silver that reached them — their cut is the reason no single merchant ever "ran" the Silk Road end to end.[5]

Han China — the agrarian mirror

Rome's structural opposite: a vast agrarian empire of some 59.6 million people at the census of 2 CE, run not on trade customs but on state monopolies of salt and iron and a land tax, and with private wealth far flatter than Rome's — the largest Han estate under a tenth of a great Roman one. The eastern anchor of the chain, and the source of the silk that moved west.[6]

[3] **Sources:** Findlay & O'Rourke (2007), ch. 1; McLaughlin (2014). **Read more:** Abu-Lughod, *Before European Hegemony* (1989), on the "relay" structure of pre-modern world systems.

[4] **Source:** McLaughlin, *The Roman Empire and the Indian Ocean* (2014).

[5] **Sources:** the brief `module_design/02_topic_2.md`; Dalrymple (Loc 1550) on Kushan control of the Hindu Kush passes.

[6] **Sources:** von Glahn, *The Economic History of China* (2016), the 2 CE census (59.6m) and Han fiscal structure; Lewis, *The Early Chinese Empires*

India — the production core

India in this period is not a single polity but a layered economy spread across the subcontinent. In the north, the Mauryan empire (c.320–185 BCE) had built South Asia's first centralised administrative-fiscal state, governed from Pataliputra — a city of perhaps 200,000 people, around twice the size of early Rome — and tied together by royal roads running from Taxila in the north-west to the Bay of Bengal port of Tamralipti. After the Mauryas fragmented, economic weight shifted south and west: the Satavahanas held the Deccan, while the Tamil kingdoms of the far south — Chera, Chola and Pandya — commanded the Malabar pepper coast that Rome most wanted. To the north-west, the Kushans straddled the Hindu Kush and the overland road to China.[7]

As everywhere in this period, the base of the economy was agriculture and the village. What set India apart was the unusually developed commercial and craft sector built on top of it — and, in particular, the two export industries that would define the subcontinent for the next eighteen centuries: cotton textiles and pepper. The Malabar coast was the world's principal source of pepper for roughly fifteen centuries, a spice so woven into Roman cooking that around four-fifths of the 478 recipes in the cookbook of Apicius call for it; the Latin and Greek words for pepper and ginger are themselves loans from Tamil. Production and long-distance exchange were organised through merchant guilds (the *śreṇi*) and, distinctively, through Buddhist monasteries placed on the trade routes, which took perpetual endowments (*akshaya-nivi*) and lent at interest — proto-financial institutions endowed by the merchant class, as the donative inscriptions at Nasik and Karle still record.[8]

India's monetary arrangements tell the centre-of-gravity story in miniature. Indian states minted their own coin — punch-marked silver in the north, local issues in the south — but they treated incoming Roman gold and silver less as foreign currency to be spent

(p.115), on the estate comparison.

than as bullion to be valued by weight and, increasingly, melted and re-struck. The incentive was structural: gold stood at roughly 10 to 1 against silver in India and on the Kushan and Saka frontier, against about 12 to 1 in Rome, so a given weight of silver carried east bought something like a fifth more goods. That arbitrage pulled Roman metal in and kept it. The clearest measure of how deep it ran is that the Kushan ruler Vima Kadphises struck India's first gold coinage on the weight standard of the Roman aureus — the empire's own bullion, re-minted as the subcontinent's coin.[9]

Externally, India was the fulcrum of the whole system — the hinge where its maritime and overland arms met. Its trade with Rome ran through three coasts: Barygaza in Gujarat to the north-west, Muziris and Nelkynda on the Malabar pepper coast, and the Coromandel ports facing the Bay of Bengal. The same merchants and the same monsoon knowledge also worked the leg eastward to Southeast Asia and Han China — the third side of the Rome–India–China triangle, and the part of the system least visible through Roman eyes. The balance of this trade ran heavily in India's favour, because Rome wanted Indian pepper, cotton, gems and ivory while India wanted little that Rome made, so the exchange settled in precious metal. So lopsided was it that, on Pliny's telling, Indian goods resold in Rome for up to a hundred times their price at source, and the Pandya and Chera kings of the south judged the relationship worth cultivating with embassies to Rome. India was, in short, where the system's most valued goods were made and where its bullion came to rest: the production core of the first world system.[10]

The smaller players — Aksum, Arabia, Alexandria

The rim that funnelled the trade: the Red Sea kingdom of Aksum, the incense kingdoms of Arabia, and Alexandria — Rome's great western emporium and the gateway to the monsoon route.[11]

How we know

Our evidence for this system is thin, indirect, and skewed. It comes mainly from three sources: merchant **handbooks** (above all the *Periplus of the Erythraean Sea*); **coin hoards**, especially the Roman coins that pile up in southern India; and a single, astonishing commercial document, the **Muziris papyrus**, recording the financing and valuation of one ship's cargo. The problem is that almost all of it is written *through Roman eyes* — which is exactly how the trade came to be read as a Roman story. A useful corrective sits on the island of **Socotra**, in the mouth of the Gulf of Aden, where a cave preserves graffiti left by ancient sailors: of 219 inscriptions, **192 are Indic** — Brahmi and Indian names — against a handful of others. Whose ocean was it? The sailors answered the question themselves.
Sources: Casson (ed.), The Periplus Maris Erythraei *(1989); De Romanis (2020) on the Muziris papyrus;*

[10]**Sources:** Roy (p.24) on the Mauryan state; Dalrymple (Loc 713) on Pataliputra's size; Dalrymple (Loc 799) on Ashoka's roads; Roy (p.32) and Dalrymple (Loc 1550) on the southern kingdoms and the Kushan passes. **Read more:** the brief `module_design/02_topic_2.md`.

[10]**Sources:** Eaton (Loc 3659) on Malabar pepper; Dalrymple (Loc 1294, 1298) on Apicius and the Tamil loan-words; on guilds and monastic finance, Schopen (1997, 2004), Ray (1994) and Neelis (2011), with the Nasik (Ushavadata) and Karle (Bhutapala; Yavana donors) inscriptions. **Read more:** the research note `module_design/topic_02_research_scale_internal_buddhism_2026-06-18.md`.

[10]**Sources:** McLaughlin, *Indian Ocean* (Loc 4323), on the 10:1-versus-12:1 ratio and the resulting margin; Dalrymple (Loc 1634, 1636) on Vima Kadphises's gold coinage.

[10]**Sources:** Roy (p.32) on the three coasts; Dalrymple (Loc 1323, 1326, 1335, 1338) on the hundredfold markup and the southern embassies. **Read more:** the India compendium `readings/notes/kindle_compendium/10_india_longrun.md`.

[11]**Source:** McLaughlin, *The Roman Empire and the Indian Ocean* (2014).

> *Strauch (ed.),* Foreign Sailors on Socotra *(2012) on the cave graffiti. Read more: the data-survival-and-bias note,* `module_design/datasets_by_topic.md`*.*

Inside the economies: three machines, all agrarian

Before following the trade outward, step inside each economy — because the long-distance trade this chapter is about touched only a sliver of any of them. **All these economies were overwhelmingly agrarian**; the vast majority of people, everywhere, farmed, and never touched a trans-Eurasian trade good. What differed was the *machinery* on top of the farms. **Rome** was the outlier: a comparatively commercial-fiscal state, its land taxes relatively light, with customs duties supplying perhaps a third of imperial revenue. **Han China** was the mirror — an agrarian state run on state monopolies of salt and iron, its wealth far flatter (its largest private estates a fraction of Rome's greatest fortunes). **India** was deeply commercial, threaded with merchant guilds — and yet its internal economy is the one we can least measure, precisely because its record did not survive in the Roman manner. Keep this in view: when we later locate the "centre of gravity," we are locating the centre of the *connected* economy, not of total output.[12]

The period on its own terms: one rise, one fall, in five phases

The first world system had a life-cycle — a single rise and a single fall — that can be read in five phases.[13]

[12] **Sources:** McLaughlin (2014) on the customs share of Roman revenue; von Glahn (2016) on Han monopolies and wealth distribution. **Read more:** Scheidel & Friesen, "The size of the economy and the distribution of income in the Roman Empire," *Journal of Roman Studies* 99 (2009).

[13] **Source:** this five-phase framing follows the Topic 2 story spine, `module_design/topic_02_story_spine_2026-06-18.md`, built from Findlay & O'Rourke (2007) and McLaughlin (2014).

Phase 1 — Opening (c.500–30 BCE). Alexander's conquests and their Hellenistic successors prise open the channels between the Mediterranean and the East; Parthia rises astride the overland route; Qin and then Han unify China; Indian Ocean trade begins to thicken. The poles of the future system take shape.[14]

Phase 2 — the switch-on (30 BCE to the first century CE). What turned sporadic contact across the Indian Ocean into a high-volume artery was, first of all, a political event: Rome's annexation of Ptolemaic Egypt in 30 BCE. Egypt handed Rome direct control of the Red Sea coast and its India-facing ports, together with the accumulated Ptolemaic experience of the eastern trade. The Greek navigator Eudoxus of Cyzicus had opened regular sailings to India around 118 BCE, and over the following century Greco-Roman skippers learned to use the monsoon — the seasonal reversal of winds, traditionally credited to a pilot named Hippalus — to run straight across the open ocean rather than creep around its rim.[15]

The change in volume was dramatic. Where the Ptolemies had sent perhaps twenty ships a year to India, Strabo reports around 120 leaving the single Red Sea port of Myos Hormos in the early imperial period — more than a sixfold rise.[16] The infrastructure behind that figure is unusually well documented for the ancient world. Two Red Sea ports, Berenike and Myos Hormos, were the Roman points of departure; the prize destination was Muziris, the Malabar pepper emporium. A merchant's handbook, the *Periplus of the Erythraean Sea*, written around the middle of the first century CE, survives to set out the ports, cargoes and sailing seasons of the whole route — the closest thing the period gives us to a commercial manual. And the cargoes were large: a single returning pepper ship of around 200 tons could carry goods worth more than six million sesterces, enough to pay six thousand

[14] **Read more:** Cunliffe, *By Steppe, Desert, and Ocean* (2015), on the opening of the channels.

[15] **Sources:** Dalrymple (Loc 1227, 1234) on Eudoxus and the Hippalus monsoon crossing; the brief `module_design/02_topic_2.md`, §1. **Read more:** Sidebotham, *Berenike and the Ancient Maritime Spice Route* (2011).

[16] **Source:** McLaughlin, *The Roman Empire and the Indian Ocean* (2014), Loc 2106, citing Strabo, *Geography* 2.5.12.

legionaries for a year.[17]

The defining feature of the switch-on, though, was the direction in which payment flowed. Rome bought pepper, fine cottons, gems, silk and incense, and paid for them overwhelmingly in gold and silver coin, because the East wanted little that Rome made. The monetary arbitrage worked in the same direction — roughly 10 to 1 gold-to-silver in India against 12 to 1 in Rome — giving the metal a further reason to move east. And move east it did, in quantities large enough to enter the Roman imagination as a grievance: this is the period in which the eastward bullion drain that defines this chapter begins in earnest.[18] It is also the period in which pepper crossed from luxury to near-staple — the spice appears in some four-fifths of the recipes in Apicius, a reach that points to a broad market rather than only an elite one, and the empirical seed of the luxury-versus-mass-trade debate this chapter will stage.[19]

By the end of the first century CE the route was regular, mature and fiscally important — customs on the eastern trade would come to supply something like a third of imperial revenue — and the stage was set for the apogee, when all four poles of the system flourished at once.

Phase 3 — Apogee (1st–2nd c. CE). At the system's height all four poles flourish together. The clearest emblem of connection comes in **166 CE**, when an embassy claiming to come from the Roman emperor "Antun" (Antoninus or Marcus Aurelius) is recorded reaching the Han court — the two ends of Eurasia, however faintly, in direct contact.[20]

Phase 4 — The turn (c.165–200). The same routes that carried goods carried disease. The **Antonine Plague** — probably smallpox — travels westward along the trade arteries and guts the Roman and the Han worlds alike; the Han military

[17] **Sources:** Casson (ed.), *The Periplus Maris Erythraei* (1989); McLaughlin, *Indian Ocean* (Loc 2766), on the 200-ton cargo and its value.

[18] **Source:** McLaughlin, *Indian Ocean* (Loc 4323), on the arbitrage and the drain.

[19] **Sources:** Dalrymple (Loc 1294) on Apicius; on the debate, De Romanis (2020) and Cobb (2018).

[20] **Source:** the *Hou Hanshu* account, via von Glahn (2016). **Read more:** Raoul McLaughlin, *The Roman Empire and the Silk Routes* (2016).

is hollowed out. Integration's gift becomes its curse.[21]

Phase 5 — Synchronised collapse (c.220–300). The poles fall *together*. The Han dynasty ends in 220; Rome plunges into its third-century crisis; the Kushan empire fragments. That the ends of the system unravelled in near-unison is itself the strongest evidence that they had been parts of one connected whole.[22]

Reading the period: the four questions

Was the system integrating or disintegrating (Q1)? Integrating, then disintegrating — and the *synchrony* of the five phases is the proof that it was one system. The poles rose together and fell together. But the very fact that the Antonine Plague could travel from one end to the other teaches the first of this book's running lessons: **the routes that integrate also transmit shocks.** Integration is not costless.[23]

Which routes carried the integration (Q2)? Both sea and land, as **complements**, not rivals. The **sea** carried bulk — pepper, gems, textiles — by the shipload, and it was far cheaper than land carriage. (*How much* cheaper is itself disputed: our sources support a range from roughly fivefold to thirtyfold, and the book takes the gap as a teaching point rather than papering over it — see the Debate below.) The **overland** routes carried high-value, low-bulk silk westward through a chain of Parthian and Kushan tolls, with each intermediary taking a cut, which is why no single merchant ran the route's length. And a crucial **third leg** ran directly between **India and China**, by the Bay of Bengal and the Kushan passes, bypassing the Mediterranean altogether. Picture the triangle again: three legs, and **Rome is only one corner**.[24]

[21] **Read more:** Harper, *The Fate of Rome: Climate, Disease, and the End of an Empire* (2017), ch. 3, for the case (and its critics) on the Antonine Plague's scale.

[22] **Sources:** von Glahn (2016) on the fall of Han; standard accounts of the third-century crisis. **Read more:** the synchrony argument in Findlay & O'Rourke (2007), ch. 1.

[23] **Read more:** on "integration breeds fragility" as a recurring mechanism, see the module synthesis, `module_design/11_synthesis.md`, Q1.

[24] **Sources:** Dalrymple (2024) and McLaughlin (2014) on relative carriage costs (the two ends of the 5×–30× range); on the India–China leg, Sen,

! Follow the money

Which flow did the actual integrating (Q3)? Treat bullion as a **good**, and it points straight to the answer. Roman precious metal flowed **east** and stayed there. *Why* east? Because the relative price of gold to silver differed across the system — gold was relatively cheaper in Rome, silver relatively dearer in India — so silver shipped east bought more. Arbitrage, in a word. The metal moved **toward the production centre**, because that is where the goods the rest of the world wanted were made and where the silver came to rest, re-minted by Kushan and Saka rulers. To grasp the scale of a single voyage: the cargo of one returning Indiaman, as valued in the **Muziris papyrus**, was worth a fortune — on one reckoning enough to pay roughly eleven thousand legionaries for a year. And the traders who carried it were, despite the Roman tint of our sources, **substantially Indian**. Follow the bullion, and the workshop sits in Asia.
Sources: De Romanis (2020) on the Muziris papyrus valuation; McLaughlin (2014) on the gold–silver arbitrage; Strauch (2012) on Indic trader presence. Read more: this is the first instalment of the book's running bullion tracer (see the Introduction, "the bullion tracer, and why it flips").

What, then, was Europe's role (Q4)? Rome was a **wealthy consumer** whose fiscal base rested on a flow of eastern goods and the customs they generated — a flow it did not control and could not produce. Powerful, rich, militarily dominant in its own sphere; but in the *economic* relay of Eurasia, the structurally dependent western terminus. Behind the whole cycle stood the **forces**: a benign climate that helped the apogee along, and then disease that took it away.[25]

Buddhism, Diplomacy, and Trade (2003). **Read more:** Diocletian's *Edict on Maximum Prices* (301 CE) as the anchor for land-vs-sea cost ratios.

[25] **Read more:** Harper, *The Fate of Rome* (2017), for the climate-and-disease framing of Rome's trajectory (and Scheidel's and others' cautions about over-reading it).

The verdict: where was the centre?

The first world system had **no single political centre** — it was a multipolar relay. But by **value of production and by the direction of bullion, Asia was its productive core**, and Rome its structurally dependent consumer. That is the centre-of-gravity verdict for this period, and it is well triangulated; what remains genuinely contested is the *scale* of the eastern trade, not its direction. (*Confidence: moderate-to-good.*)[26]

It is essential to size the claim **honestly**, and this is where the chapter sharpens rather than inflates the spine. The eastern luxury trade was, on a reasonable estimate, only around **5% of Roman output** — large for its reach, but marginal within an economy whose real weight was internal grain and oil. (Indeed the single largest sea trade in the Roman world was not the pepper run at all, but the **Mediterranean grain and oil** that fed the cities.) So "centre of gravity" here does **not** mean the centre of total output, which was overwhelmingly agrarian everywhere. It marks the centre of the **connected** economy — where the world's *valued* goods were made and where its bullion pooled. And the "success" we are measuring is **aggregate scale**, not income per head. The per-capita switch — the thing that will eventually make European dominance possible — comes much later, in Chapter . (This is the **quantity to quality** thread, set running here.)[27]

> ⚠ The debate: a luxury haemorrhage, or high-volume mass trade?
>
> Was the Indo-Roman trade a thin trickle of ruinously expensive luxuries for a tiny elite — the "haemorrhage" of Pliny's complaint — or a **high-volume commerce** in goods like pepper and ordinary textiles that reached well down the social scale? The older view emphasised

[26] **Source:** centre-of-gravity ledger, `module_design/centre_of_gravity_ledger.md`, Topic 2 entry.

[27] **Sources:** the ~5%-of-GDP estimate follows the scale discussion in `module_design/topic_02_research_scale_internal_buddhism_2026-06-18.md`, drawing on Scheidel and on Temin, *The Roman Market Economy* (2013). **Read more:** on the quantity-vs-quality distinction across the book, see the Introduction.

luxury and elite consumption; a revisionist line (De Romanis; Cobb) reads the **Muziris papyrus** and the scale of the pepper trade as evidence of something much closer to a mass market. The dispute matters because it changes how *economically* significant the system was, and it is bound up with a second, unresolved quarrel — the value of the *Hermapollon* cargo recorded in that papyrus, which different scholars reconstruct very differently (a full value around 9 million sesterces, on one reading, with duty above 2 million). This book does **not** quietly pick a winner. The gap in the cost-of-carriage figures (roughly 5× versus 30×) and the *Hermapollon* reconstruction are presented as exactly what they are: open questions where the evidence underdetermines the answer.
Sources: De Romanis, The Indo-Roman Pepper Trade and the Muziris Papyrus *(2020); Cobb,* Rome and the Indian Ocean Trade from Augustus to the Early Third Century CE *(2018).*

Data exhibit — Roman coin hoards in southern India. The densest physical trace of the bullion drain is the distribution of Roman silver and gold coins found in hoards across the Indian peninsula, clustering in the Tamil south and trailing off north and east. Plotted against the *Periplus*'s port list and the Muziris papyrus, the hoards turn a literary claim ("metal flowed east") into a mappable pattern. *Caveats:* hoards record **deposition and non-recovery**, not flow — a coin buried and never retrieved is over-represented relative to one that kept circulating; and find-spots reflect where modern archaeology has looked. *What you could do with this:* combine published hoard catalogues with port locations to ask whether coin density tracks the *Periplus*'s emporia, and test how sensitive the picture is to the deposition bias.
Sources: published Roman-coin-hoard catalogues for South Asia (e.g. Turner, 1989; updated finds); Casson (1989) for the Periplus *ports. See the per-topic dataset entry in* `module_design/datasets_by_topic.md`*, Topic 2.*

Threads forward

This chapter set three of the book's threads running. The **centre-of-gravity** verdict gets its first real entry (Asia the productive core of a multipolar relay; moderate-to-good confidence). The **quantity-versus-quality** thread is introduced with a warning we will need for every "thriving" empire to come: aggregate scale is not income per head. And **integration breeds fragility** got its first demonstration in the Antonine Plague.[28]

One actor deserves a closing look, because it carries us into Chapter . In India, **Buddhism** and long-distance trade reinforced one another. Congenial to mobile merchants because it sat lightly on the caste-ritual barriers that constrained others, it was endowed by merchant guilds — the donative inscriptions at **Nasik** (the financier Ushavadata) and **Karle** (the *setthi* Bhutapala; named **Yavana**, i.e. Greek/Roman, donors) record the flow of commercial wealth into the monasteries. Sited on the trade routes, those monasteries became **proto-financial institutions**: they took perpetual endowments (*akshaya-nivi*) and lent at interest, supplying credit and trust along the roads — and carrying Buddhism itself on toward China. (Note the careful claim: *proto-banks*, not "the first banks," and we drop the once-common story that a Brahmanical "sea taboo" drove merchants into Buddhism — the evidence does not support the causal link.) Hold on to that pairing of **religion and trade**, because the next chapter opens a new order built around two faiths — Islam and a Tang-Chinese world — in which religion again lubricates the exchange.[29]

Classic research: the foundations

The modern debate sits on top of a century-old argument among economic historians about *what kind of economy* antiquity even had — an argument worth knowing, because this chapter's "one market or a chain of empires?" question is its direct descendant.

[28] **Read more:** the six cross-cutting threads are set out in the Introduction; their per-chapter payoffs are tracked in `module_design/11_synthesis.md`.

[29] **Sources:** Schopen, *Bones, Stones, and Buddhist Monks* (1997) and *Buddhist Monks and Business Matters* (2004) on monastic endowments and lending; Ray, *The Winds of Change* (1994) and Neelis, *Early Buddhist Transmission and Trade Networks* (2011) on monasteries and routes; Gernet, *Buddhism in Chinese Society* (1995). **Read more:** the evidence note `module_design/topic_02_research_scale_internal_buddhism_2026-06-18.md`.

- **Rostovtzeff (1926)**, *The Social and Economic History of the Roman Empire* — the **"modernist"** foundation: a Roman world of vigorous trade, investment and a quasi-bourgeois commercial class. Internet Archive
- **Tenney Frank, ed. (1933–40)**, *An Economic Survey of Ancient Rome* (5 vols) — the foundational compilation that first assembled the scattered quantitative evidence.
- **Polanyi, Arensberg & Pearson, eds (1957)**, *Trade and Market in the Early Empires* — the **"substantivist"** view: ancient exchange was *embedded* in social and political institutions, not a self-regulating market. The intellectual root of "relay of empires, not one market."
- **Finley (1973)**, *The Ancient Economy* — the **"primitivist"** classic that dominated for a generation: status, not markets, organised ancient economic life; long-distance trade was marginal. The paradigm the quantitative turn was built to test. UC Press
- **Hopkins (1980)**, "Taxes and Trade in the Roman Empire (200 B.C.–A.D. 400)," *Journal of Roman Studies* 70: 101–125 — the **seminal quantitative model** linking taxation, money and trade; the hinge between Finley's qualitative world and modern cliometrics. JSTOR
- *On this chapter's exact subject* — **Warmington (1928)**, *The Commerce between the Roman Empire and India*, the classic synthesis of the Indo-Roman trade (Internet Archive); and **Wheeler, Ghosh & Krishna Deva (1946)**, "Arikamedu," *Ancient India* 2 — the excavation that put the trade on a material footing.

At the research frontier: recent cliometric work

A quantitative ("cliometric") literature now tries to put **numbers** on the ancient economy — output, living standards, market integration — and it is moving quickly. The list below reflects a sweep of **NBER, CEPR and RePEc/IDEAS** (plus the leading archaeology/ancient-history venues), recent first. *One honest result of the sweep:* NBER surfaced no ancient-economy working paper; the economics-discipline action is in **CEPR / *Review of Economic Studies*** and **JEP / EHR**, while the field's empirical depth sits in **OXREP / JRS / *Science Advances***.

- **Flückiger, Hornung, Larch, Ludwig & Mees (2022)**, "Roman Transport Network Connectivity and Economic Integration," *Review of Economic Studies* 89(2): 774–810 — the headline economics-discipline paper: uses spatially disaggregated excavated-ceramics data to show Roman transport connectivity drove **market integration** (and still shapes integration two millennia on). Squarely on this chapter's **Q2 (channels)**. Circulated as a CEPR Discussion Paper. DOI · RePEc · CEPR column
- **Ortman, Lobo, Lodwick, Wiseman, Bulik, Harbison & Bettencourt (2024)**, "Identification and measurement of intensive economic growth in a Roman imperial province," *Science Advances* 10(27): eadk5517 — settlement-scaling methods find *modest per-capita* (intensive) growth in Roman Britain over four centuries. A direct test for this book's **quantity-vs-quality** thread. DOI
- **Kessler & Temin (2007)**, "The organization of the grain trade in the early Roman Empire," *Economic History Review* 60(2): 313–332 — uses grain-price gradients to argue for an **integrated** Mediterranean market; a key plank of the "modernist/market" revival.
- **Temin (2006)**, "The Economy of the Early Roman Empire," *Journal of Economic Perspectives* 20(1): 133–151 — the accessible economics-discipline statement of the market case, expanded in his *The Roman Market Economy* (2013). DOI
- **De Romanis (2020)**, *The Indo-Roman Pepper Trade and the Muziris Papyrus* (OUP) — the definitive quantitative reconstruction behind this chapter's central document; the empirical core of the luxury-vs-mass-trade debate.
- **Bowman & Wilson, eds (2009–)**, *Quantifying the Roman Economy* and the ongoing **Oxford Roman Economy Project** — the standing programme turning proxies (shipwrecks, amphorae, coins, mining pollution) into economic indicators; the methodological backbone of the field.
- **Scheidel & Friesen (2009)**, "The size of the economy and the distribution of income in the Roman Empire," *Journal of Roman Studies* 99: 61–91 — the benchmark

Roman-GDP estimate behind this chapter's "~5% of output" sizing.

- **Milanović, Lindert & Williamson (2011)**, "Pre-industrial inequality," *Economic Journal* 121: 255–272 — places Rome in a comparative inequality frame (the "inequality possibility frontier").

> **ℹ Keeping this current (verification gate)**
>
> Sweep run 2026-06-25 across NBER, CEPR, RePEc/IDEAS, Crossref and JSTOR; every entry above was citation-verified (DOI/stable URL). The deepest *ancient*-economy cliometrics still sits in archaeology/ancient-history venues more than in NBER/CEPR (which skew modern). Refresh before publication and **never list an unverified working paper**. The "Classic research" + "research frontier" pair recurs in every chapter; later chapters (7–10) will be far more NBER/CEPR-heavy.

Questions for consideration

Essay / exam style — each rewards the four-questions toolkit and the chapter's live debate, not recall.

1. *"Rome was the consumer, Asia the centre."* Assess this claim using the eastward bullion drain.
2. Was the first Eurasian system **one market** or **a chain of empires**? Use the synchrony of its rise and fall, and the role of the middlemen, in your answer.
3. *"The Indo-Roman trade was a luxury sideshow, not a structural feature of either economy."* How would you test this — and how far do the data (the ~5%-of-output estimate; the Muziris papyrus) let you?
4. How much explanatory weight should **climate and disease** carry in the rise and fall of the first world system? Discuss with reference to the Antonine Plague and the third-century crisis.

5. *“Absence of evidence is not evidence of absence.”* Using the Socotra graffiti and the Roman coin hoards, discuss how the **survival and bias** of the sources shapes what we can claim about who ran this trade.

> Cross-cutting questions (collected at the end of the book)
>
> The book closes with a bank of questions that span several chapters. Two that this chapter feeds: - **Bullion as tracer.** Compare the eastward Roman drain (this chapter) with the eastward early-modern silver flow (Chapter): in what sense does “follow the bullion” locate the same kind of centre in both, and where does the analogy break down? - **Quantity vs quality.** Across Chapter , Chapter and Chapter , when is it legitimate to call a “thriving” empire *rich*, and when only *big*? Build your answer from aggregate-vs-per-capita evidence.

Data exercise

```
# Indo-Roman coin hoards vs Periplus ports (datasets_by_topic.m
# 1. Load a published catalogue of Roman coin hoards in South A
# 2. Load Periplus emporia coordinates.
# 3. Map hoard density; test whether it tracks the port list.
# 4. Discuss the deposition/non-recovery bias before drawing in
```

Key data

Figure	Value	Source (to wire to `references.bib`)
Roman drain east (rhetorical)	~100m sesterces/yr	Pliny, via Dalrymple 2024
Customs share of Roman revenue	~1/3	McLaughlin 2014
Sea vs land carriage cost	~5×–30× cheaper (disputed)	Dalrymple 2024 / McLaughlin 2014
Monsoon-era shipping increase	~6×	Strabo, via McLaughlin 2014

Figure	Value	Source (to wire to `references.bib`)
Eastern trade as share of Roman output	~5%	Scheidel; research note
Hermapollon cargo value	~9m sesterces full; duty >2m (disputed)	De Romanis 2020
Socotra graffiti	192 of 219 Indic	Strauch 2012
Direct Rome–Han contact	"Antun" embassy, 166 CE	*Hou Hanshu*, via von Glahn 2016
Pataliputra population	~200,000	Dalrymple 2024

The two figures flagged "disputed" are presented in the Debate, not silently resolved. Formal citation keys are wired during the Phase-3 bibliography consolidation.

Further reading

- **Core:** Findlay & O'Rourke, *Power and Plenty* (2007), ch. 1; McLaughlin, *The Roman Empire and the Indian Ocean* (2014).
- **Supplementary:** Dalrymple, *The Golden Road* (2024); von Glahn, *The Economic History of China* (2016), on the Han anchor; Casson (ed.), *The Periplus Maris Erythraei* (1989, primary).
- **The debate:** De Romanis (2020) and Cobb (2018) on the Muziris papyrus and luxury-vs-mass trade; on Buddhism and commerce, Ray (1994), Schopen (1997, 2004), Neelis (2011).

II — The ocean at its height

Maritime apogee: Baghdad and Guangzhou, not Aachen

The premodern Indian Ocean at its height — and Europe at its most peripheral.

> **Draft pending**
>
> **Period:** c.500–1000. This chapter is a stub. Draft it from the standardized template (`_chapter-template.qmd`) using its story spine (`teaching/ecu4413_world_econ/module_design/topic_03_story_spine_2026-06-19.md`), the matching topic brief, the centre-of-gravity ledger, the data compendium, and `datasets_by_topic.md`.

The high-medieval boom and the Mongol climax

The system's fullest premodern extent — then the great shock.

ℹ Draft pending

Period: c.1000–1400. This chapter is a stub. Draft it from the standardized template (`_chapter-template.qmd`) using its story spine (`teaching/ecu4413_world_econ/module_design/topic_04_story_spine_2026-06-19.md`), the matching topic brief, the centre-of-gravity ledger, the data compendium, and `datasets_by_topic.md`.

III — The world becomes one — and stays Asian

The world becomes one — and stays Asian

> *Three oceans connect; silver welds the globe; the sink is still in Asia.*

Draft pending

Period: c.1400–1600. This chapter is a stub. Draft it from the standardized template (`_chapter-template.qmd`) using its story spine (`teaching/ecu4413_world_econ/module_design/topic_05_story_spine_2026-06-19.md`), the matching topic brief, the centre-of-gravity ledger, the data compendium, and `datasets_by_topic.md`.

Companies, coercion and the contest of the oceans

Europe gains organised, armed leverage — while Asia stays the workshop.

> **ℹ Draft pending**
>
> **Period:** c.1600–1760. This chapter is a stub. Draft it from the standardized template (`_chapter-template.qmd`) using its story spine (`teaching/ecu4413_world_econ/module_design/topic_06_story_spine_2026-06-19.md`), the matching topic brief, the centre-of-gravity ledger, the data compendium, and `datasets_by_topic.md`.

IV — The great reversal

The Great Divergence: the centre moves west

The hinge of the whole story — and its most open question. That the centre moved is not in doubt; when it moved, and why, is the book's central debate.

Follow one thing: a bolt of cotton cloth

Follow a single bolt of cotton. In 1700 the finest cloth in the world was Indian — calico and chintz woven on hand looms in Bengal and on the Coromandel coast, so coveted in Europe that Britain and France **banned** it to protect their own weavers. A century and a half later the flow had reversed: cheap machine-spun cloth poured *out* of Lancashire and flooded the world, including India, whose hand-loom weavers it undercut in their own bazaars. The same product — cotton cloth — that had once measured Asian supremacy now measured its eclipse. This chapter is the story of that reversal, because the cloth carries the whole argument: the moment the centre of gravity of the world economy moved from Asia to the North Atlantic.[1]

Where we are on the arc

Through Chapter , the centre of gravity still sat in Asia: India made perhaps a quarter of the world's manufactures as late as 1750, and Europe was the armed, silver-bearing contestant, not yet the workshop. This chapter is **the hinge**. By 1850 the centre has moved to the North Atlantic — that much

[1] **Sources:** Findlay & O'Rourke, *Power and Plenty* (2007), ch. 5; Parthasarathi, *Why Europe Grew Rich and Asia Did Not* (2011), on the calico bans. **Read more:** Riello, *Cotton: The Fabric that Made the Modern World* (2013); Beckert, *Empire of Cotton* (2014).

is uncontested. What is genuinely open, and what this chapter stages rather than settles, is *when* the move began and *why* — and whether Europe was ever structurally ahead before it happened. This is the topic on which the book's whole "bounded anomaly" verdict turns, so we return to weigh it in the Chapter .[2]

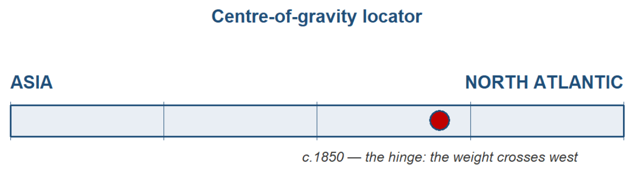

Figure 1: Centre-of-gravity locator (c.1850): the hinge — the weight crosses from Asia to the North Atlantic.

The stage and the cast

The cast carries over from Chapter , but the roles reverse. **Britain** turns from a calico *importer* and armed contestant into the mechanised workshop of the world. **India** turns from the world's cotton workshop into a deindustrialising periphery — its share of world manufacturing collapsing from roughly a quarter in 1750 to under 3% by 1880, the single headline reversal of the whole book. **Qing China** is the great parity comparator: the lower Yangzi delta tracked Europe's leading regions until about 1700 before falling behind. Two newer western actors join: the **United States South**, whose enslaved-labour cotton plantations became the raw-fibre engine of Lancashire; and **West Africa**, whose story is the monetary mirror-image — a deindustrialisation worked through *money* rather than machines. And, as the foil that keeps the question honest, **continental Europe**: if it was simply "Europe," why Britain and not France?[3]

[2] **Sources:** the centre-of-gravity ledger, `module_design/centre_of_gravity_ledger.md`, Topic 7 (the pivotal entry); the module synthesis, `module_design/11_synthesis.md`, Q4. **Read more:** Pomeranz, *The Great Divergence* (2000), the book that named the debate.

[3] **Sources:** Bairoch (1982) and Findlay & O'Rourke (2007), ch. 5, on the manufacturing-share reversal; Broadberry, Guan & Li (2018) on the

Dramatis personae

- **Britain** (*since Chapter : transformed*) — from armed contestant and calico importer to the mechanised "workshop of the world."
- **India** (*present every chapter; since Chapter : reversed*) — from the world's cotton workshop to a deindustrialising periphery; the chapter's central case.
- **Qing China** (*since Chapter : changed*) — the parity comparator that shadowed Europe's leaders to ~1700, then diverged.
- **The US South / the Atlantic slave-cotton complex** (*new*) — the enslaved-labour raw-fibre engine; the New World "ghost acres" feeding mills the land of Britain could not.
- **West Africa** (*new role*) — deindustrialisation by monetary divergence (the parallel case).
- **Continental Europe (France)** — the foil for "why Britain, specifically?"

How we know — here the data *is* the debate

In every other chapter the evidence is sparse and we squeeze what we can from it. Here the problem inverts: there is a *lot* of evidence, and the disagreement is about **which numbers to trust**. The divergence is fought with rival reconstructions — Bairoch's manufacturing shares, Broadberry's historical national accounts, Allen's real wages — and we have far richer series for Britain and Europe than for Asia. So "when did the divergence begin?" is partly an artefact of *where we have data*: the book's running **data-bias** warning bites hardest precisely here, on the chapter that decides the whole argument. Two honesty checks set the tone. First, the Industrial Revolution itself was *slower* than legend — the Crafts–Harley reconstruction cut the headline growth rates sharply. Second, even the best Asian series get revised: the leading estimate of historical Chi-

Yangzi parity; Beckert (2014) on the US South; Green, *A Fistful of Shells* (2019), on West Africa. **Read more:** Parthasarathi (2011) on the India case.

> nese GDP was publicly challenged and restated. Treat every number in this chapter as a contested estimate, not a fact.
> *Sources: Crafts & Harley, "Output growth and the British Industrial Revolution: a restatement," Economic History Review 45 (1992); Broadberry, Guan & Li (2018) and the subsequent "Further Concerns"/"Restatement" exchange in the Journal of Economic History. Read more: the data-survival-and-bias note,* `module_design/datasets_by_topic.md`.

Inside the economies: three engines, and Britain's is the new kind

Before the narrative, look inside — because the divergence is, at bottom, a story about three different economic engines. **Britain** was, on the standard account, a high-wage, cheap-coal economy: expensive labour and cheap energy made it profitable to *adopt* and to *invent* labour-saving, coal-burning machines. The **lower Yangzi** was the opposite — an "involutionary" economy of extraordinary labour intensity, absorbing ever more work-hours per acre for diminishing returns (by one comparison, labour intensity some eight times England's, for output only nine times higher). **India** was a low-wage handloom proto-industry: genuinely competitive in world markets, but competitive *on the basis of cheap labour*, not mechanisation. The factor-price contrast is the cleanest single mechanism to teach — and, as we will see, the most contested.[4]

[4] **Sources:** Allen, *The British Industrial Revolution in Global Perspective* (2009), on factor prices; on Yangzi involution, the discussion in von Glahn (2016) and Pomeranz (2000); on Indian proto-industry, Parthasarathi (2011). **Read more:** the contest over Allen's mechanism — Kelly, Mokyr & Ó Gráda (2023), in the research-frontier section below.

The period on its own terms: how cotton turned around, 1750–1850

The reversal runs in five phases, from Indian primacy to a North-Atlantic workshop.[5]

Phase 1 (c.1700–1750) — The eve: India the workshop, parity at the top. Indian calicoes flood European markets; Britain legislates against them (1700, 1721) to shelter home weavers. The Yangzi delta still tracks Europe's leading economies. No one is yet pulling away.[6]

Phase 2 (1750s–1780s) — Britain mechanises cotton. A cascade of textile inventions — Kay's flying shuttle (1733), Hargreaves's jenny, Arkwright's water frame (1769), Crompton's mule (1779) — meets Newcomen's and then Watt's steam engine, and coal turns the wheels. The effect on productivity is staggering: the operative-hours needed to process 100 lb of cotton fall from upwards of 50,000 by hand to around 2,000 on the early mule.[7]

Phase 3 (1780s–1815) — The reversal gathers; the Atlantic engine fires. Raw cotton pours from the US South after Whitney's gin (1793), grown by enslaved labour on New-World land that functioned as "ghost acres" Britain did not have at home. British raw-cotton imports leap from roughly 3,600 tonnes (1775) toward 27,000 tonnes (1800); cotton output triples. Britain flips from calico importer to the world's dominant cloth exporter.[8]

Phase 4 (1815–1840s) — India deindustrialises; Lancashire floods the world. Productivity keeps climbing — operative-hours per 100 lb reach about 135 by 1825 — and by 1815 some 60% of British cotton textiles are exported. This is a productivity-driven terms-of-trade shock: Lancashire cloth

[5] **Source:** the five-phase framing follows the Topic 7 story spine, `module_design/topic_07_story_spine_2026-06-19.md`, built from Findlay & O'Rourke (2007), ch. 5, and Riello (2013).

[6] **Sources:** Parthasarathi (2011) on the calico acts; Broadberry, Guan & Li (2018) on Yangzi parity. **Read more:** Riello (2013), part II.

[7] **Sources:** Findlay & O'Rourke (2007), ch. 5, for the operative-hours series; standard accounts of the textile inventions. **Read more:** Allen (2009), chs 6–8, on why these inventions paid in Britain specifically.

[8] **Sources:** Beckert (2014) on the slave-cotton complex; Pomeranz (2000) on "ghost acres"; Findlay & O'Rourke (2007) for the import figures. **Read more:** Beckert (2014), chs 3–5.

undercuts Indian weavers *in India*. And coercion has teeth — the Permanent Settlement (1793), Company exports financed from Bengal's land revenue, the end of the East India Company's trade monopoly (India 1813, China 1833) — "free trade," on British terms.[9]

Phase 5 (1840s–1850s) — The centre has moved; China is cracked open. British raw-cotton imports run from ~220,000 tonnes (1840) to ~680,000 tonnes (c.1860); India's share of world manufacturing falls below a tenth. China's parity ends in the Daoguang Depression, with net silver flowing *out* from the late 1820s; the Opium War (1839–42) and the Treaty of Nanjing force it open. By 1850 the centre sits in the North Atlantic.[10]

Reading the period: the four questions

Direction (Q1) is the subtle one. The divergence is *not* a break in integration — the world keeps integrating across these decades — but a **reversal of relative position within an integrating system**. That carries the chapter's distinctive fragility lesson: integration can *hollow out* a core, not only lift it. India was not cut off from the world economy; it was remade by it, on disadvantageous terms.[11]

Channels (Q2): the integrating circuit is now the cotton circuit — raw fibre *in*, manufactured cloth *out*, on terms Britain sets. Its raw-material leg is the Atlantic slave-sugar-cotton complex; the old overland channel is now marginal. The geography of the world economy has been re-plumbed around an Atlantic-and-Indian-Ocean manufacturing core.[12]

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[9] **Sources:** Findlay & O'Rourke (2007), ch. 5; Parthasarathi (2011) and Beckert (2014) on coercion; on the deindustrialisation-as-terms-of-trade-shock mechanism, Clingingsmith & Williamson. **Read more:** the deindustrialisation debate in the *Debate* box below.

[10] **Sources:** Findlay & O'Rourke (2007), ch. 5, for the cotton-import series; von Glahn (2016) on the Daoguang Depression and silver outflow. **Read more:** on the China shock, Chapter (the first global economy on colonial terms).

[11] **Sources:** O'Rourke & Williamson on integration-with-divergence; the module synthesis, `module_design/11_synthesis.md`, Q1. **Read more:** Williamson, *Trade and Poverty: When the Third World Fell Behind* (2011).

[12] **Source:** Beckert (2014) on the Atlantic raw-material leg. **Read more:**

! Follow the money — the silver thread pays off

Modes (Q3): goods lead now — the cotton reversal is the headline flow — and bullion is a *good* on its way to becoming money (the role-change completes in Chapter). But here the book's **"follow the silver"** thread, running since Chapter , finally pays off. Europe's centuries as the monetary loser — shipping silver east to buy Asian goods it could not match — turn out to have been the making of it: Europe **converted its monetary "loss" into armed trade, finance and navies.** Up to 90% of British state revenue went on war; the European edge was fiscal-military power and organised violence as much as cheaper manufactures. West Africa is the parallel run in reverse: deindustrialisation by *monetary* divergence — the macuta sliding from 500 to 40 reais, some 30 billion cowries absorbed — money, not machines, doing the hollowing-out.
Sources: the silver thread note, `module_design/silver_monetization_thread_2026-06-16.md`*; Green, A Fistful of Shells (2019), on West Africa; on the fiscal-military state, O'Brien and Findlay & O'Rourke (2007). Read more: the Introduction's "bullion tracer, and why it flips".*

Europe (Q4) is the question this chapter exists to test — and the one period where the answer is genuinely "yes, Europe pulls ahead." *Why Britain?* There is no settled answer, and the honest move is to teach the disagreement. One quartet of accounts explains the *adoption* of the new technology: **Allen** (factor prices — high wages plus cheap coal made machines pay); **Pomeranz** (resources — coal near at hand, plus New-World ghost acres); **Mokyr** (useful knowledge — an Industrial Enlightenment); **Beckert** (war capitalism — empire and slavery). A second quartet locates a *deeper* European peculiarity well before 1750: **Mokyr** (a culture of growth), **Scheidel** (the productive consequences of political fragmentation), **Acemoglu & Robinson** (inclusive institutions), **Rubin** (religion and the relation of rulers to clerics). The Parthasarathi twist keeps the spine honest: the divergence emerged *from* India's cotton dominance

Findlay & O'Rourke (2007), ch. 5.

— British mechanisation was in part a response to Indian competition — not in spite of it.[13]

Forces. Technology leads — mechanisation, steam, coal — and the debate is really about what made invention *and adoption* happen *here*. Energy is the tell: English daily energy use rose from roughly 33,000 to 99,000 kcal per head, a jump sustained only by coal. Climate enters too, as a supply shock in parts of the Indian story.[14]

The verdict: where was the centre?

That the centre of gravity moves west by 1850 is **uncontested**. *When* and *why* it moved is **genuinely open** — and on the answer hangs the strength of this book's spine. If the divergence was **late and contingent** (Pomeranz, Allen — parity until ~1750, then a coal-and-colonies accident), the "European dominance as anomaly" framing is *stronger*. If it was **early and structural** (Broadberry — India behind Britain by the 17th century, China diverging from ~1700), the framing is *weaker*. The reconciliation worth teaching is **"late in outcome, earlier in roots"**: the aggregate per-head gap opens late, but specific seeds — institutional, fiscal, demographic — run earlier. (*Confidence: high on the move; the* why *is the debate — the ledger's pivotal, deliberately-unresolved line, adjudicated at the* Chapter .)[15]

Then **size it honestly**. The Industrial Revolution was gradual: the Crafts–Harley reconstruction puts British growth at roughly 0.33% a year, with output expanding about 6.5-fold between 1700 and 1860. Cotton was the lead sector, not

[13] **Sources:** Allen (2009); Pomeranz (2000); Mokyr, *A Culture of Growth* (2016); Beckert (2014); Scheidel, *Escape from Rome* (2019); Acemoglu & Robinson, *Why Nations Fail* (2012); Parthasarathi (2011). **Read more:** all are reviewed in the research-frontier section below; the author's own Allen/Pomeranz notes slot in here.

[14] **Sources:** Scheidel and Wrigley on the coal-energy transition; Findlay & O'Rourke (2007), ch. 5. **Read more:** Wrigley, *Energy and the English Industrial Revolution* (2010).

[15] **Sources:** centre-of-gravity ledger, `module_design/centre_of_gravity_ledger.md`, Topic 7; Broadberry, Custodis & Gupta (2015) and Broadberry, Guan & Li (2018) for the early-divergence case; Pomeranz (2000) and Allen (2009) for the late case. **Read more:** the *Debate* box below.

the whole economy; the dramatic headline hides a slow aggregate transformation. And note *what kind* of change this was. The per-capita divergence opens around 1700 — *during* the Asian empires' aggregate apogee — which is why this is the chapter where the book's **quantity to quality** thread finally resolves: **the Great Divergence *is* the switch.** "Success" stops meaning aggregate scale and starts meaning sustained income per head. By 1700 China sat at perhaps 55% of the European leader; India fell from over 60% of Britain's GDP per head in 1600 to under 15% by 1871; a day's wage around 1700 bought roughly four subsistence baskets in London but only one in Beijing, Canton or Edo.[16]

> ⚠ The debate: late and contingent, or early and structural?
>
> This is *the* debate of the module. The **California school** (Pomeranz, Allen) argues for surprising **parity** between the most advanced regions of Europe and Asia as late as ~1750, with the divergence then opening fast on coal and colonies — late, contingent, and so a *stronger* case for European dominance as an anomaly. The **historical-national-accounts** camp (Broadberry and colleagues) reconstructs GDP series showing the gap opening **earlier and more structurally** — India behind Britain by the seventeenth century, China sliding from ~1700 — a *weaker* anomaly reading. Bound up with it are two further quarrels: **why Britain** (Allen's factor-prices vs Mokyr's knowledge vs Pomeranz's resources vs Beckert's empire — and the newer claim of Kelly, Mokyr & Ó Gráda that it was *low* wages plus mechanical skill, not high wages, that drove adoption); and **deindustrialisation** — was India's handloom sector *destroyed* (Williamson, Beckert) or *reorganised and partly resilient* (Roy, Harnetty)? The book does not resolve any of these. It insists only that **measurement is the debate**: which series you trust decides which story you tell.

[16]**Sources:** Crafts & Harley (1992) for the growth reconstruction; Broadberry, Guan & Li (2018) and Broadberry, Custodis & Gupta (2015) for the GDP ratios; Allen and Findlay & O'Rourke (2007) for the real-wage baskets. **Read more:** the quantity-vs-quality thread is set up in Chapter and closed in Chapter .

> *Sources: Pomeranz, The Great Divergence (2000); Allen (2009); Broadberry, Custodis & Gupta (2015); Broadberry, Guan & Li (2018); Kelly, Mokyr & Ó Gráda (2023); on deindustrialisation, Roy and Harnetty vs Williamson/Beckert. All cited in full below.*

Data exhibit — follow the cloth: India's deindustrialisation in numbers. Lay three series side by side, 1600–1880: India's **share of world manufacturing** (~24.5% in 1750 to under 3% by 1880, Bairoch); India's **GDP per head relative to Britain** (>60% in 1600 to <15% by 1871, Broadberry–Custodis–Gupta); and British **cotton-cloth exports** (surging from ~1815). Read together they turn "India was deindustrialised" from a slogan into a measured reversal — and show *when* each curve turns, which is the empirical heart of the timing debate. *Caveats:* the manufacturing-share and GDP series are **reconstructed estimates** with wide error bands, far better-grounded for Britain than for India; the manufacturing-share denominator (world output) is itself estimated. *What you could do with this:* plot the three series, mark the inflection years, and test how the "when did it begin?" answer moves as you swap Bairoch's shares for Broadberry's GDP — i.e. show that measurement *is* the debate.
Sources: Bairoch, "International Industrialization Levels from 1750 to 1980," Journal of European Economic History 11 (1982); Broadberry, Custodis & Gupta (2015); Findlay & O'Rourke (2007), ch. 5. See the per-topic dataset entry in `module_design/datasets_by_topic.md`, Topic 7.

Threads forward

This is the chapter where the book's threads converge. The **centre-of-gravity** spine reaches its hinge (the move west by 1850 — high confidence; the *why* left open for the Chapter). The **quantity to quality** thread resolves: the divergence *is* the switch from aggregate scale to income per head. And the **"follow the silver"** thread pays off — Europe's monetary "loss" converted into power.[17]

Hand-off to Chapter : by 1850 the centre has moved, and in the next chapter this newly Euro-centred order *globalises* — the first price-integrated world economy, built on steam, the Suez Canal and indentured labour, and resting on the very Indian-Ocean periphery it now governs. The divergence made the dominance; the next chapter shows what the

[17] **Source:** the threads are tracked in `module_design/11_synthesis.md`. **Read more:** the Introduction sets out all six.

dominance was *for*.[18]

Classic research: the foundations

Long before "the Great Divergence" was named, economic historians argued about *why Europe* and *why the Industrial Revolution* — and the chapter's modern debate descends directly from them.

- **Weber (1905)**, *The Protestant Ethic and the Spirit of Capitalism* — the original culture-and-institutions account of European economic dynamism; the ancestor of Mokyr's and Rubin's arguments.
- **Gerschenkron (1962)**, *Economic Backwardness in Historical Perspective* — relative backwardness and the varieties of industrialisation; reframed "catching up" as a process.
- **Landes (1969)**, *The Unbound Prometheus*, and **Landes (1998)**, *The Wealth and Poverty of Nations* — the influential (Eurocentric) technological-superiority account the California school was written against.
- **E. L. Jones (1981)**, *The European Miracle* — the environment-and-politics case for Europe's distinctiveness; the foil for Pomeranz.
- **Bairoch (1982)**, "International Industrialization Levels from 1750 to 1980," *Journal of European Economic History* 11: 269–333 — the manufacturing-share series (the ~24.5% to 3% India figures) that still anchors the descriptive picture.
- **Crafts & Harley (1992)**, "Output growth and the British Industrial Revolution: a restatement," *Economic History Review* 45(4): 703–730 — the reconstruction that showed the IR was *slower* than legend; the basis for this chapter's "size it honestly."

[18] **Source:** the arc, `module_design/module_outline_2026-06-16.md`, §6. **Read more:** Chapter .

At the research frontier: recent cliometric work

The Great Divergence is one of the most active quantitative fields in economic history, and — unlike the ancient chapters — it is **NBER/CEPR-central**. A sweep of NBER, CEPR and RePEc, with the defining books, recent first:

- **Kelly, Mokyr & Ó Gráda (2023)**, "The Mechanics of the Industrial Revolution," *Journal of Political Economy* 131(1): 59–94 (CEPR DP 14884) — using county-level English data, argues industrialisation went with *low* wages and *high mechanical skills*, not Allen's high wages, and that coal proximity and literacy don't predict it. A direct, current challenge to the factor-price thesis. DOI · CEPR
- **Broadberry, Guan & Li (2018)**, "China, Europe, and the Great Divergence: A Study in Historical National Accounting, 980–1850," *Journal of Economic History* 78(4): 955–1000 — the GDP reconstruction placing the China–Europe divergence *earlier* than the California school allows; see also the subsequent "Further Concerns"/"Restatement" exchange (the field correcting itself in public). RePEc
- **Allen (2015)**, "The high wage economy and the industrial revolution: a restatement," *Economic History Review* 68(1): 1–22 — Allen defending the factor-price account against its critics; read alongside Kelly–Mokyr–Ó Gráda as the live exchange. Wiley
- **Broadberry, Custodis & Gupta (2015)**, "India and the great divergence: An Anglo-Indian comparison of GDP per capita, 1600–1871," *Explorations in Economic History* 55: 58–75 — the India series behind this chapter's headline ratios. RePEc
- **Mokyr (2016)**, *A Culture of Growth* (Princeton) — the Industrial-Enlightenment / culture account, now the leading "deep roots" alternative to factor-price stories.
- **Allen (2009)**, *The British Industrial Revolution in Global Perspective* (CUP) — the high-wage/cheap-coal thesis in full; the most-cited single mechanism.
- **Parthasarathi (2011)**, *Why Europe Grew Rich and Asia Did Not* (CUP) — the India-centred account: mechanisation as a *response* to Indian competition.
- **Pomeranz (2000)**, *The Great Divergence* (Princeton) —

the book that defined the debate (parity to ~1800; coal + New-World ghost acres).

Keeping this current (verification gate)

Sweep run 2026-06-25 across NBER, CEPR, RePEc/IDEAS and Crossref; every entry above was citation-verified (DOI/RePEc/publisher URL). This is a fast-moving field — refresh before publication and **never list an unverified working paper**. *Standing build note:* integrate the author's own (non-Kindle) Allen and Pomeranz reading notes into the Q4 "why Britain" paragraphs.

Questions for consideration

Essay / exam style — each rewards the four-questions toolkit and the live debate, not recall.

1. Was the Great Divergence **late and contingent** or **early and structural**? Set out what evidence would decide it, and how far the available series actually let you.
2. *"India was deindustrialised, not outcompeted."* Discuss, distinguishing market forces from coercion, and the "destroyed" from the "reorganised" readings.
3. Why **Britain**, and not France, China or India? Assess at least two of the four "why-the-West" accounts (Allen, Pomeranz, Mokyr, Beckert) against the evidence.
4. *"The Great Divergence is the moment 'success' changed meaning."* Explain, using the quantity-versus-quality distinction and the per-capita series.
5. *"Here the data is the debate."* Using the rival GDP and real-wage reconstructions, show how the answer to "when did the divergence begin?" depends on which series you trust — and on where we have data at all.

Cross-cutting questions (collected at the end of the book)

Those this chapter feeds: - **Quantity vs quality.** Across Chapter , Chapter and Chapter , when is a "thriving" empire *rich* and when only *big*? Build the answer from aggregate-vs-per-capita evidence. - **How bounded is the anomaly?** Holding Chapter 's timing debate against Chapter 's partial return, how strong a version of "European dominance as anomaly" can the evidence support?

Data exercise

```
# The Great Divergence: when did it begin? (datasets_by_topic.m
# 1. Load Broadberry et al. GDP-per-capita series (Britain, Ind
# 2. Load Bairoch manufacturing-share series and Allen real-wag
# 3. Plot each; mark the inflection year implied by each measur
# 4. Show how the "start date" of the divergence shifts with th
#    measurement IS the debate. Note the wider error bands on t
```

Key data

Figure	Value	Source (to wire to `references.bib`)
India share of world manufacturing	~24.5% (1750) to <3% (1880)	Bairoch 1982 / Findlay–O'Rourke 2007
India GDP/cap vs Britain	>60% (1600) to <15% (1871)	Broadberry, Custodis & Gupta 2015
China GDP/cap vs European leader	~55% by 1700	Broadberry, Guan & Li 2018
Cotton operative-hours per 100 lb	>50,000 (hand) to ~2,000 (mule 1779) to 135 (1825)	Findlay–O'Rourke 2007

Figure	Value	Source (to wire to `references.bib`)
British raw-cotton imports	3,600 t (1775) to 27,000 t (1800) to ~680,000 t (c.1860)	Findlay–O'Rourke 2007
English energy use per head/day	33,000 to 99,000 kcal	Scheidel; Wrigley
IR aggregate growth	~0.33%/yr; output ~6.5× (1700–1860)	Crafts–Harley 1992
Real wage c.1700 (baskets/day)	~4 London vs ~1 Beijing/Canton/Edo	Allen; Findlay–O'Rourke 2007
West Africa monetary divergence	macuta 500 to 40 reais; ~30bn cowries	Green 2019

GDP and manufacturing-share figures are reconstructed estimates with wide error bands (better for Britain than for Asia) — presented as the substance of the debate, not as settled facts.

Further reading

- **Core:** Findlay & O'Rourke, *Power and Plenty* (2007), ch. 5; Pomeranz, *The Great Divergence* (2000); Allen, *The British Industrial Revolution in Global Perspective* (2009).
- **Supplementary:** Parthasarathi, *Why Europe Grew Rich and Asia Did Not* (2011); Riello, *Cotton* (2013); Beckert, *Empire of Cotton* (2014); Broadberry, Custodis & Gupta (2015).
- **The debate:** Kelly, Mokyr & Ó Gráda (2023) vs Allen (2015) on the high-wage thesis; Broadberry et al. (2018) vs the California school on timing; Roy/Harnetty vs Williamson/Beckert on deindustrialisation.

The first global economy: integration on colonial terms

Steam, Suez and empire — peak Western dominance, resting on the ocean it governed.

> **Draft pending**
>
> **Period:** c.1850–1914. This chapter is a stub. Draft it from the standardized template (`_chapter-template.qmd`) using its story spine (`teaching/ecu4413_world_econ/module_design/topic_08_story_spine_2026-06-20.md`), the matching topic brief, the centre-of-gravity ledger, the data compendium, and `datasets_by_topic.md`.

V — Break and return

Disintegration: the mirror of the global economy

Deglobalisation — and how the cracking of the core empowered the periphery.

ℹ Draft pending

Period: 1914–1945. This chapter is a stub. Draft it from the standardized template (`_chapter-template.qmd`) using its story spine (`teaching/ecu4413_world_econ/module_design/topic_09_story_spine_2026-06-20.md`), the matching topic brief, the centre-of-gravity ledger, the data compendium, and `datasets_by_topic.md`.

The Asian rebalancing: the arc resolves

The centre returns east — partially, and within a Western-built order.

Draft pending

Period: 1945–2010s. This chapter is a stub. Draft it from the standardized template (`_chapter-template.qmd`) using its story spine (`teaching/ecu4413_world_econ/module_design/topic_10_story_spine_2026-06-20.md`), the matching topic brief, the centre-of-gravity ledger, the data compendium, and `datasets_by_topic.md`.

Conclusion: How strong is the spine?

Now that the evidence is in, re-weigh the bounded-anomaly claim.

> **i** Draft pending
>
> Draft from `11_synthesis.md` + the end-of-module revisit: re-weigh the 'bounded anomaly'; the 1945-vs-1970s break; aggregate-vs-per-capita 'return'; what the very long run teaches.

Cross-cutting questions

The bank of essay/exam questions that span several chapters lives here; each chapter points forward to the ones it feeds. *Build by collecting every chapter's "Cross-cutting questions" call-out, grouped by thread — centre-of-gravity, quantity to quality, follow-the-silver, channels, forces, data-and-bias.*

> **i** Draft pending — seeded from Chapter and Chapter
>
> 1. **Bullion as tracer.** Compare the eastward Roman drain (Chapter) with the eastward early-modern silver flow (Chapter): in what sense does "follow the bullion" locate the same kind of centre in both, and where does the analogy break down?
> 2. **Quantity vs quality.** Across Chapter , Chapter and Chapter , when is it legitimate to call a "thriving" empire *rich*, and when only *big*? Build the answer from aggregate-vs-per-capita evidence.
> 3. **How bounded is the anomaly?** Holding Chap-

ter 's timing debate (late-&-contingent vs early-&-structural) against Chapter 's partial return, how strong a version of "European dominance as anomaly" can the evidence support? (*The spine's final adjudication.*)

Datasets catalogue

> **Draft pending**
>
> From `teaching/ecu4413_world_econ/module_design/datasets_by_topic.md` (near camera-ready): per-topic datasets with student-ready / illustrative / specialist flags, the four-questions mapping, and what a student could do with each.

A guide to the sources and their biases

> **ℹ Draft pending**
>
> The 'data survival and bias' thread, consolidated: record-keeping conventions, climate/ preservation, 'absence is not absence', and 'through whose eyes'. From the module's data-survival note and each chapter's 'How we know'.

The centre-of-gravity ledger

> **Draft pending**
>
> The arc at a glance, as one consolidated table, from `teaching/ecu4413_world_econ/module_design/centre_of_gravity_ledger.md` — each chapter's verdict and confidence level in one place.

References